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Sensitivity of surface water and groundwater contributions to
streamflow in a tropical glacierized basin under climate change
scenariosCaroline Aubry-Wake^{1,*} , Gavin McNamara^{2,3}, Lauren D Somers⁴ , Jeffrey M McKenzie² ,
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E-mail: caroline.aubrywake@gmail.com**Keywords:** glacier, groundwater, Cordillera Blanca, cold region hydrological model, mountain hydrology, climate changeSupplementary material for this article is available [online](#)

Abstract

While mountain water faces threats posed by climate change, particularly in snow-dominated and glacierized systems, the role of groundwater (GW) in sustaining streamflow in these systems remains elusive. Changing mountain headwaters, marked by reduced snowpacks, retreating glaciers, shifting precipitation patterns, and rising temperatures, pose a crucial question: what is the resilience of streamflow in these mountains, and what role does GW play in this resilience? This is particularly uncertain in tropical high mountains where the seasonality of precipitation and glacier melt govern streamflow generation. A glacio-hydrological model was created using the Cold Regions Hydrological Modelling platform to investigate cryosphere-surface water–GW interactions in the Quilcayhuanca Basin, in Peru's Cordillera Blanca. The model was forced by *in-situ* meteorological observations and parameterized using numerous data sources and process-based studies in the basin. Model results show that during the dry season, 37% of streamflow is generated from GW discharge, increasing to 56% during the lowest flows. Evapotranspiration is the largest mass flux from the basin at the peak of the dry season. Precipitation, temperature, and glacier change scenarios were used to assess the sensitivity of basin hydrology to climate change and glacier retreat. In a warmer, wetter, and nearly deglaciated future, Quilcayhuanca basin streamflow is expected to decrease by 4%–19% annually, with a larger volumetric change in overland and vadose zone flow than in GW flow. The range in values is more closely linked to uncertainty in precipitation change than temperature change. Despite a strong reduction in snow and ice contribution to streamflow with warming and deglaciation, the concomitant increase in precipitation can limit the changes in streamflow and GW flow, showcasing the resilience of the system to shifts in climate and glacier cover.

1. Introduction

Groundwater (GW) in high mountain regions is thought to be crucial in buffering the impact of climate change and variability on water resources (Hayashi 2020). While mountain glaciers (when present) are critical for sustaining streamflow, there

are significant knowledge gaps regarding interactions between glaciers, mountain hydrology/hydrogeology, and downstream water supply (van Tiel *et al* 2024). Specifically, while glacier melt contributions to streamflow and surface water are complex but well-studied, high-mountain GW processes have received scant but increasing attention over recent decades

(Somers and McKenzie 2020). GW has the potential to store and then release water across seasons, although the processes controlling the transfer of water to and from GW reservoirs are poorly understood in mountain regions. In many mountain ranges around the world, limited monitoring infrastructure and sparse data collection hinder a comprehensive understanding of hydrological processes, and in particular GW processes (Somers and McKenzie 2020, Thornton *et al* 2022).

Research from the Cordillera Blanca, Peru, has shown that the Rio Santa's water supply during the dry season relies heavily on glacial meltwater and stored GW (Bury *et al* 2011). For the upper Rio Santa watershed, glacial meltwater contributes approximately 10%–20% of the total annual discharge, with the potential to exceed 40% during the dry season (Mark *et al* 2005). In contrast, GW can contribute between 24% and 80% of the dry-season flows in the individual sub-basins of the Rio Santa (Baraer *et al* 2015). These findings highlight the substantial role of both glaciers and GW in sustaining streamflow in the Rio Santa basin.

Several previous studies have shown that both climate change and glacier recession are impacting water resources in the Peruvian Andes. Baraer *et al* (2012) show that seven of nine rivers studied in the Cordillera Blanca have passed a threshold glacier volume (peak water) and now exhibit declining streamflow. Hydrological models of varying complexities have been used to project future climate change impacts and generally agree that there will be a decrease in the dry season streamflow as glaciers continue to retreat. Representation of GW in these models ranges from very simple (e.g. Juen *et al* 2007) to three-dimensional flow (e.g. Somers *et al* 2019). Presently, not one hydrological model adequately captures physical hydrological processes from the glaciers to the deep subsurface. Fully integrated models, which can capture both the complex snow and ice dynamics in the mountain terrain and GW flow are difficult to adequately parameterize and calibrate due to the uncertainty in the subsurface representation and can be computationally expensive (Carroll *et al* 2020, Thornton *et al* 2022). A simplified approach is therefore desirable to expand the applicability of hydrological projections in mountains where both glaciers and GW are substantial contributors to streamflow, and where insights on future mountain water resources are needed to adapt and mitigate ongoing changes (van Tiel *et al* 2024). The combination of existing studies about the importance of both glacier and GW contribution to streamflow in the Rio Santa basin, and the associated understanding of the hydrological system, provides a unique opportunity to develop a middle-ground complexity hydrological modelling framework to estimate how both

GW and streamflow can respond to glacier retreat and climate change.

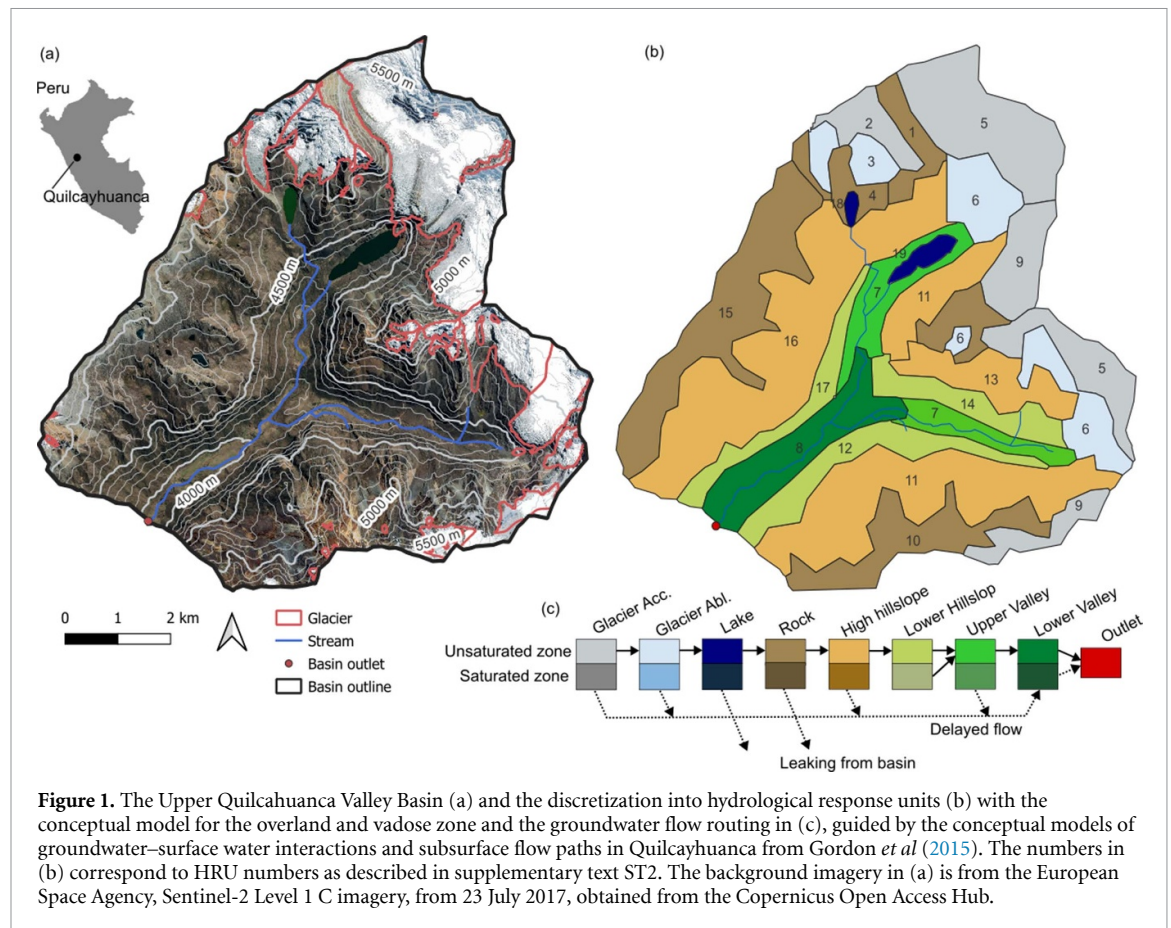
The goals of this study are to showcase how, when based upon a physical understanding of GW processes obtained from extensive field-based investigations, a simplified representation of the surface water–GW processes can capture key groundwater flow behaviour. Then, using this modelling framework, GW–surface water interactions are assessed in a tropical glacierized basin. Finally, the sensitivity of a tropical, glacierized, mountain basin's hydrological regime is examined through the application of climate change and glacier retreat scenarios.

2. Methods

2.1. Study area and hydrogeological context

The study basin is the upper Quilcayhuanca Valley, situated in the Cordillera Blanca, Peru ($9^{\circ}27' \text{ S}$, $77^{\circ}22' \text{ W}$). The Cordillera Blanca, a subrange of the northern Peruvian Andes, has the highest concentration of tropical glaciers worldwide (Kaser *et al* 1990). The region experiences distinct wet and dry seasons, with more than 80% of the annual precipitation occurring between October and May (Baraer *et al* 2009). Typically, river discharge in the area peaks from February to March due to the South American Monsoon System (Arias *et al* 2021). The temperature range is characteristic of tropical regions, where the daily temperature fluctuation surpasses the monthly variation throughout the year (Kaser *et al* 1990).

The upper Quilcayhuanca Valley Basin has an area of 67.4 km², which was 26% covered by glaciers (2015), including the Cuchillacocha glacier (RGI ID G282650E09398SRGI). The elevation in the upper Quilcayhuanca Valley Basin ranges from 3900 m. a.s.l. to 6157 m. a.s.l., with an average elevation of 4800 m. a.s.l. (figure 1(a)). Two automated weather stations (figure 1(a)) are in the basin. The higher elevation station is situated near Laguna Cuchillacocha ($9^{\circ}25' \text{ S}$, $77^{\circ}20' \text{ W}$, 4642 m. a.s.l.) at the headwaters of the northern arm of the valley. The lower elevation station, known as Casa De Agua (CDA), is located on the valley floor ($9^{\circ}27' \text{ S}$, $77^{\circ}22' \text{ W}$) alongside the Quilcay River at an elevation of 3924 m a.s.l. and serves as the outlet of the basin. Both weather stations are equipped to collect hourly hydrometeorological observations, including air temperature ($^{\circ}\text{C}$), precipitation (mm), relative humidity (%), wind speed (ms^{-1}), and incoming solar radiation (W m^{-2}) since July 2013, but with significant gaps in their record. At the lower elevation station (CDA), water level data is available for the period July 2015 to June 2019, with a gap between January and October 2016 (as seen in figure 3(a)). Water level data is obtained at a gauging station near a channel cut in the stream, in a large meadow where water was previously rerouted



for agricultural purposes. The rating curve for this location was built using nine points collected at different discharge levels, as presented in Mateo *et al* (2022). Six GW observation wells are located in the lower pampa near the CDA station, providing hourly measurements since 2013. More information on the hydrometeorological data collection can be found in Mateo *et al* (2022) and is shown in supplementary text ST1.

The bottom of the Quilcahuanca Valley is a long, topographically flat, high-elevation valley-bottom meadow referred to as a ‘pampa’. The valley floor contains four distinct depositional layers, consisting of shallow, dry, organic-rich soils underlain by water-saturated sandy, organic-rich soils. These layers are further underlain by an approximately 3 m-thick, effectively impermeable glacial-lacustrine sediment layer, which hydrologically isolates the more permeable layer beneath. Below the clay layer, there are coarse-grained gravels from colluvial deposits, intermixed with glacial outwash till and diamicton, referred to as buried talus (Glas *et al* 2018).

The talus deposits are estimated to be at a minimum of 15 m thick, and the geometric mean hydraulic conductivity of these talus layers is $2.1 \times 10^{-5} \text{ m s}^{-1}$ (Chavez 2013). Based on their relatively high hydraulic conductivities and lateral extent, these buried talus layers are classified as confined aquifers (Glas *et al* 2018) and artesian conditions

have been observed for these buried talus deposits. Seasonal recharge to these aquifers is thought to be primarily from wet-season flows travelling down the steep rock valley walls and channelled through the talus slopes (Gordon *et al* 2015, Glas *et al* 2018).

GW–surface water exchange has been relatively well studied in the Quilcahuanca Valley. GW makes up approximately 40% of the Quilca River dry-season flows where the river exits the valley (Baraer *et al* 2015). GW upwells contribute to streamflow through direct discharge and extensive springs on the valley floor. Two cross-valley moraines have been identified as sites of considerable GW–surface water exchange through tracer studies (Gordon *et al* 2015, Somers *et al* 2016). Here, steeper slopes and coarser sediments facilitate enhanced exchange of GW and surface water.

This combination of multi-year shallow piezometers and multiple GW-focused processes-based studies, a unique wealth of knowledge of GW processes for a high-elevation, glacierized basin, provide a robust understanding of the GW flow path in the basin, which is used to guide the application of a hydrological model to the area.

2.2. Modelling framework: the cold regions hydrological modelling platform

The object-oriented, flexible Cold Regions Hydrological Modelling Platform (Pomeroy *et al*

2007, 2022) was utilized to develop a coupled, process-based glacio-hydrological model based on modules describing the physical processes of precipitation phase change, snow redistribution by wind and gravity, accumulation of snow and rainfall at the surface, firnification of snow into glacier ice, energy fluxes from radiation and turbulent transfer resulting in snowmelt, firn melt, icemelt, sublimation and evapotranspiration, infiltration into soils and to dynamical GW reserves, runoff generation via surface and subsurface flows, and exfiltration from GW to streamflow. This purpose-built model provided a framework to investigate the interactions between surface water and GW and assess the sensitivity of the Upper Quilcayhuanca basin to climate and glacier change. Here, the wet (October to April) and dry season (May to September) are compared. Peak dry season was in August, which typically has the lowest streamflow and peak wet season was December, when streamflow is highest.

The flexibility of the CRHM modelling framework allows us to include the unique field-based knowledge of the Quilcayhuanca basin in the model application. Specifically, the flexibility in the routing module allows the knowledge of the GW flow path, delays and storages, to be included in the hydrological model. GW-specific parameters, such as hydraulic conductivities, can be directly transferred from these studies (e.g. Chavez 2013) to the area of the basin where they apply. In addition to this, the spatial discretization of the model, through hydrological response units, is designed to include the interactions between surface water and GW, such as the differentiation between an upper valley bottom and a lower valley bottom, as these two regions are divided by steeper short sections that allow surface water–GW mixing (Gordon *et al* 2015, Somers *et al* 2016), and a steep, lower hillslope section, where coarse sediments such as talus deposits, enhance GW recharge and direct this GW to the lower valley aquifer (Glas *et al* 2018). The flexibility of the model allows the creation of falsification simulations to assess the response of the basin to varying conditions. In this case, considering the focus on GW parameterization, a falsification simulation which ignores the GW storage and delays by setting the corresponding parameters to their minimal values is conducted to test the importance of the appropriate GW representation in this system. An additional falsification simulation with the absence of glacier ice is designed to assess how it affects the streamflow composition (surface + overland flow and GW).

The model spatial discretization, hydrometeorological inputs, parameterization and evaluation are briefly described below, with more details found in the corresponding supplementary materials.

2.2.1. Spatial discretization

The spatial discretization in CRHM is based on hydrological response units (HRUs), zones with common hydrological characteristics and properties such as topography, vegetation cover, soil properties, glacier coverage and hydrometeorological attributes. Land cover classification was performed using a combination of field-based observations and satellite imagery. In this CRHM application, the HRUs were designed to first take into account land cover type: glacier accumulation zone, glacier ablation zone, high-elevation rock, high-elevation hillslope, low-elevation hillslope, upper valley, and lower valley, and then to capture the existing knowledge of the GW flow paths and the connectivity between the surface water and GW systems, as discussed in Gordon *et al* (2015), Somers *et al* (2016) and Glas *et al* (2018). Details on the selection of the HRUs are found in supplementary text ST2.

2.2.2. Hydrometeorological inputs preparation and basin-wide distribution

The model was forced with hourly observations of air temperature, relative humidity, shortwave radiation, wind speed, and precipitation recorded at the CDA station (3924 m. a.s.l.) between 26 June 2014, and 6 April 2020. To fill data gaps, bias-corrected observations from the Cuchillacocha weather station located 7 km upstream of the CDA station at an elevation of 4642 m, and measurements from the Llaganuco Valley approximately 50 km southwest of the study basin were combined.

Temperature was distributed across the basin following monthly temperature gradients between -0.66°C to -0.79°C per 100 m elevation in the wet and dry seasons, respectively, obtained from measurements in the Llaganuco basin at elevations between 3955 to 4561 m. a.s.l. Precipitation gradients were based on the measurements from the two stations in the study basin, Cuchillacocha and CDA (figure 1(a)). The same precipitation gradient was used for all seasons, as there were no clear seasonal variations noticeable in the precipitation gradient. The precipitation phase is based on the wet bulb temperature, with a temperature threshold of -3°C for precipitation to fall fully as snow, and 0°C for precipitation to fall as rain. Wind flow acceleration and deceleration over the Quilcayhuanca complex terrain, an important component to capture blowing snow processes and turbulent transfer calculations, were simulated following a linearized turbulence model (Walmsley *et al* 1986). Radiation was adjusted for self-shading and slope-aspect following the formulations of Garnier and Ohmura (1970), but it was not corrected for shading from surrounding terrain. More information about the data used for observations can be found in Mateo *et al* (2022) and information about

data gap-filling, elevation gradients for precipitation and temperature and resulting basin-wide meteorological forcings is found in Supplementary Text ST1.

2.2.3. Hydrological processes parameterization

Most parameters in the CRHM modules have a physical meaning and can be obtained from field sites or remote sensing observations (for example, the temperature elevation gradient or vegetation height). When parameters specific to the site are unavailable, they can be transferred from compatible hydrological environments described in the scientific literature. Parameters for streamflow routing, however, do not have a direct physical meaning, and cannot be directly measured or transferred between studies, as they depend on HRU size, connectivity and distribution. These routing parameters are therefore first defined based on the conceptual understanding of the surface water–GW interactions in the basin, such as the knowledge that water percolating to the GW reservoir in the hillslopes travels to the main valley aquifer and is stored there seasonally and were subsequently manually calibrated to better match the timing of measured streamflow. This approach avoids the need to calibrate physically identifiable parameters from streamflow. More details on model structure, parameterization, process description and sensitivity analysis can be found in the supplementary text ST3. For a more complete description of the process representation in the model, see Aubry-Wake *et al* (2022), Pradhananga and Pomeroy (2022), Pomeroy *et al* (2022). For analysis, the surface runoff and soil layer runoff are grouped and called overland and vadose flow (OV + VAD) respectively, and the runoff from the GW storage is referred to as GW.

The routing parameterization was designed to combine process understanding from existing literature and manual calibration. For most HRUs, the leakage from the GW storage was routed to the valley bottom GW HRU, and the surface streamflow was routed to the adjacent downstream HRU (figure 1(c)). To represent GW leakage outside of the modelled basin (i.e. GW that discharges to the surface somewhere downstream or becomes mountain block recharge) certain HRUs were selected where GW leakage was assumed to flow 100% outside the model (i.e. GW leakage was removed from the model). These few HRUs included the lake HRUs, a subset of the high rock, glacier ablation zone HRUs, and the upper valley HRU. These HRUs were selected because they are relatively high elevation and therefore likely have the deepest flow paths and are more likely to discharge outside the basin. These assumptions allow some GW to leave the basin through the subsurface corresponding to GW flow that does not resurface within the basin and exits the basin in the shallow subsurface or is lost to deeper GW flow paths. The true GW leakage outside the basin is unknown at the study site,

as in most mountain study sites. However, allowing some leakage outside the basin is more realistic than forcing all recharge to resurface based on studies in other mountain regions (e.g. Gleeson and Manning 2008, Gonzalez-Duque *et al* 2024). Furthermore, a complete sensitivity analysis of the GW routing structure is performed in ST3.2.

2.2.4. Model evaluation

The model was compared to measured streamflow at the basin outlet and GW storage change in the valley bottom. For GW storage change, the water level in the six observation wells was averaged and converted to storage change using porosity based on the soil type observed in the valley and compared to the simulated GW storage in the valley bottom HRU (HRU 8). The discharge volume comparison between the overland flow plus vadose zones and GW zone contribution to streamflow was then compared to existing literature values.

To assess the performance of the glacier mass balance and melt module, the simulated melt was compared to Fyffe *et al* (2021), who also applied a physically based, energy balance melt model at five glacier sites within Peru, including Cuchillacocha Glacier, and to measured ablation from Aubry-Wake *et al* (2018), as well as glacier melt rates from literature throughout the Cordillera Blanca. Evapotranspiration rates are also compared to literature values (Hellström and Mark 2006, Wigmore 2016). This model evaluation, constrained by multiple processes and datasets is a crucial approach in glacierized mountains (van Tiel *et al* 2020). The use of a multi-year, multi-well GW level is especially unique, as these datasets are not common for high-altitude, glacierized basins (Somers and McKenzie 2020).

2.3. Sensitivity of streamflow to temperature, precipitation and glacier cover

Scenarios are designed to assess streamflow sensitivities to temperature, precipitation and glacier retreat. This approach follows the delta change factor method (López-Moreno *et al* 2017, 2020, Rasouli *et al* 2022), which can produce plausible hydroclimatological changes for the future, retaining the main covariances of forcing fields and variability in hydrometeorological processes present in historical measurements whilst minimizing computational resources.

Under RCP 4.5, the maximum daily temperature is predicted to increase by 1.9 ± 0.6 °C, and by 3.6 ± 0.8 °C under RCP 8.5 (Potter *et al* 2023), with other estimates suggesting an increase up to 5 °C by the end of the century (Bradley *et al* 2004, Solman 2013). To capture the range of temperature increase until the end of the century, temperatures were increased by 0 °C–5 °C, in increments of 1 °C.

Predictions of precipitation change are harder to assess due to the difficulty in establishing historical

trends and spatial patterns in precipitation over the Andes mountains. However, large-scale studies over South America suggest an increase in future precipitation in the wet season in the Peruvian Andes, based on CMIP6 models (Almazroui *et al* 2021). Potter *et al* (2023) found an increase in $5.8 \pm 6.3\%$ and $12.1 \pm 11.0\%$ for RCP 4.5 and 8.5. Precipitation change in complex mountain terrain from climate models is notoriously uncertain (Lutz *et al* 2016, López-Moreno *et al* 2017), and therefore, despite recent studies suggesting a likelihood of increased precipitation, precipitation was multiplied by 0.8–1.2, in increments of 0.1. In this approach, the number of precipitation events and the seasonality of the events is maintained, with only the volume changing. This is a reasonable assumption, given recent evidence that precipitation change in the future is linked to increasing precipitation on wet days, rather than an increase in the number of wet days (Potter *et al* 2023), a pattern that is reproduced with a fixed precipitation multiplicative factor but does not capture the increase in precipitation extremes.

Three glacier cover scenarios were run for glacier coverages of 0%, 2%, and 13%, compared to 26% in 2015 (figure ST2.3). These numbers, constrained by the distribution of glacier HRUs, are supported by limited literature estimates of future glacier retreat in the region. Based on a recent set of global glacier projections for every glacier on Earth, the sum of all individual glaciers in the basin by 2100 is predicted to be $8.22 \pm 4.69\%$, $3.45 \pm 3.05\%$, and $0.02 \pm 1.12\%$ from 2015 conditions for scenarios RCP 2.6, RCP 4.5 and RCP 8.5, respectively (Rounce *et al* 2023, 2022; shown in figure ST2.4 and table ST2.3 for RCP and SSP scenarios). These numbers, obtained from simulations using flowline models based on the shallow-ice approximation, are calculated for each individual glacier in the basin, producing the most up-to-date and advanced future glacier estimate for the region. Schauwecker *et al* (2017) find that under RCP 8.5, only 1% of the current area of the glacier in the Cordillera Blanca would be above the future freezing height level, a common proxy to estimate glacier mass balance in the Tropics (Rabatel *et al* 2012). Juen *et al* (2007) found a glacier area loss of 75% under the A2 scenario.

The scenarios of air temperature and precipitation change applied to basins with varying glacier cover allow direct comparison of the streamflow response as a function of glacierized areas and provide insights into the hydrological functioning of these basins. With these scenarios, it is possible to estimate the additional precipitation needed to counterbalance the impacts of rising temperatures and/or glacier retreat on streamflow and gain insights into the relationship between temperature, precipitation and glacier melt. However, it is essential to note that certain scenario combinations, such as high glacier

cover coupled with significant temperature increase and precipitation decrease, are physically improbable. More plausible scenarios are those combining glacier covers of 0% or 2% with precipitation and temperature change resembling conditions under RCP 8.5 scenarios, and combinations of 13% glacier cover and more moderate temperature and precipitation changes akin to RCP 2.6. However, due to the uncertainty in the response of high mountain weather to climate change, and the complex interactions between temperature, precipitation and glacier mass balance in this complex terrain, the range of scenarios are presented. Following this broader analysis of the sensitivity of the basin to changes, scenarios with a higher likelihood of occurring, based on the limited literature available for the area, are further analysed.

3. Results and discussion

3.1. Multivariate model evaluation

The modelled daily and weekly streamflow at the basin outlet provided reasonable agreement with the measured streamflow (table 1, figure 3(a)). A unity value of NSE indicates perfect agreement between simulations and observations while a value of zero means that the model simulations have the same explanatory power as the mean of the observations (Schaeffli and Gupta 2007). For the Kling Gupta efficiency (KGE), values greater than 0.41 indicate that a model improves upon the mean flow benchmark (Knoben *et al* 2019).

The simulated glacier melt and ET rates compared well to previous studies in the Quilcayhuanca Valley and wider region (table 2). Modelled GW storage and measured storage, assuming an effective porosity of 0.45–0.55 (fine to medium sand, silt, loamy or clayey sands) to convert the measured water level to changes in storage, showed strong agreement for the timing of the peaks (wet season) and troughs (dry season) and storage volume, with a Pearson correlation coefficient between the average measured GW level and modelled GW storage of 0.88 (figure 2). Overall, these comparisons to streamflow discharge, glacier ablation and mass balance, evapotranspiration and GW storage suggest that the model can predict the components of the hydrological cycle in this basin sufficiently well for use in diagnostic investigations of climate change.

3.2. Flow composition and hydrological fluxes

The streamflow at the basin outlet is primarily generated by runoff of overland and vadose zone flow during both the wet and dry seasons (figure 3(a), table 3), with higher proportional GW contributions during the dry season. The CRHM GW routing parameterization, allowing water draining from the GW reservoir for specific HRUs (lakes, high elevation rock and

Table 1. Comparison of measured and modelled streamflow metrics with the Nash Sutcliffe efficiency parameter (NSE), the Kling Gupta efficiency (KGE), the root mean square error (RMSE), the mean absolute error (MAE), the coefficient of determination (R^2) and the mean bias (MB) for the 2015–2019 period.

Streamflow	NSE	KGE	RMSE	MAE	R^2	MB
Daily	0.69	0.83	0.59	0.40	0.70	−2%
Weekly	0.77	0.87	0.48	0.32	0.80	−2%

Table 2. Comparison of simulations to previous glacier and ET studies. Previous studies are within the Quilcayhuanca basin unless otherwise noted.

Attribute	Simulated value in present study	Previous study	Value in previous study
Surface mass balance July 2014–July 2016	−8.4, −4.3 and −2.6 m w.e. for glacier toes at 4893, 5020 and 5152 m. a.s.l., respectively.	Fyffe <i>et al</i> (2021)	−9 m w.e. at 4821 m.a.s.l.
Average glacier mass loss rate	−3.9, −1.7, −0.6, +0.04 and +0.4 m w.e. yr ^{−1} at 4893, 5020, 5152, 5251 and 5495 m.a.s.l., respectively.	Rabatel <i>et al</i> (2013)	−1.2 m w.e. yr ^{−1} for tropical Andean glaciers with a maximum elevation below 5400 m a.s.l.
Ablation zone glacier surface melt from June 23–10 July 2014	1.7, 4.6, and 12.8 cm for ablation zones at 5020, 4893, and 5152 m a.s.l., respectively	Aubry-Wake <i>et al</i> (2018)	12 ± 3 cm (mean ± standard deviation), range = 9–19 cm at an elevation of 4821 m a.s.l.
Annual average ET rate	1.78 mm d ^{−1}	Hellström and Mark (2006)	1.33 mm d ^{−1} in nearby Llanganuco Valley
		Wigmore (2016)	1–3 mm d ^{−1} , with higher variability in wet season than dry season
Dry season ET rate	1.81 mm d ^{−1}	Hellström and Mark (2006)	0.03 mm d ^{−1} in nearby Llanganuco Valley
Wet season ET rate	1.76 mm d ^{−1}	Hellström and Mark (2006)	2.63 mm d ^{−1} in nearby Llanganuco Valley

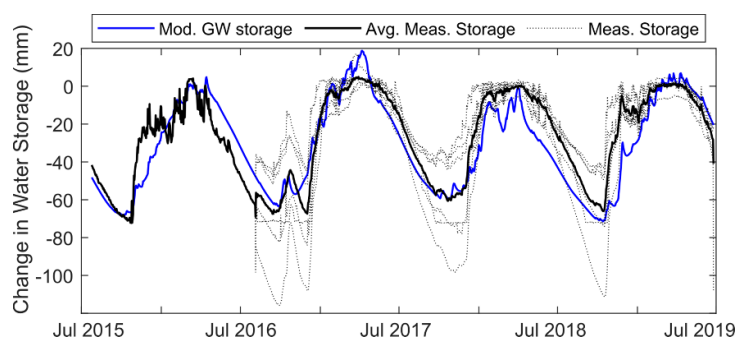


Figure 2. Change in groundwater storage in valley bottom aquifer, comparing the storage for HRU 8 and measured storage based on an effective porosity of 0.055. Both data sets are centred on their average monthly maximum, which is calculated as the highest average value among all months over the entire time series. The measured water level from the piezometer and CRHM groundwater storage can be seen in figure S1.

some glacier HRUs, see figure ST3.2., text ST2.3 and text ST3.2 for discussion) to leave the basin instead of being routed to the valley bottom aquifer, resulted in 3% of the basin yield leaking from the basin. This water could include both mountain block recharge and water travelling in the shallow subsurface without

returning to the stream network within the simulated watershed. This approach was selected to capture the likelihood of a leaky basin (Fan 2019, Markovich *et al* 2019), based on existing literature about mountain bedrock GW flowpaths (Gleeson and Manning 2008, Gonzalez-Duque *et al* 2024). Applying GW

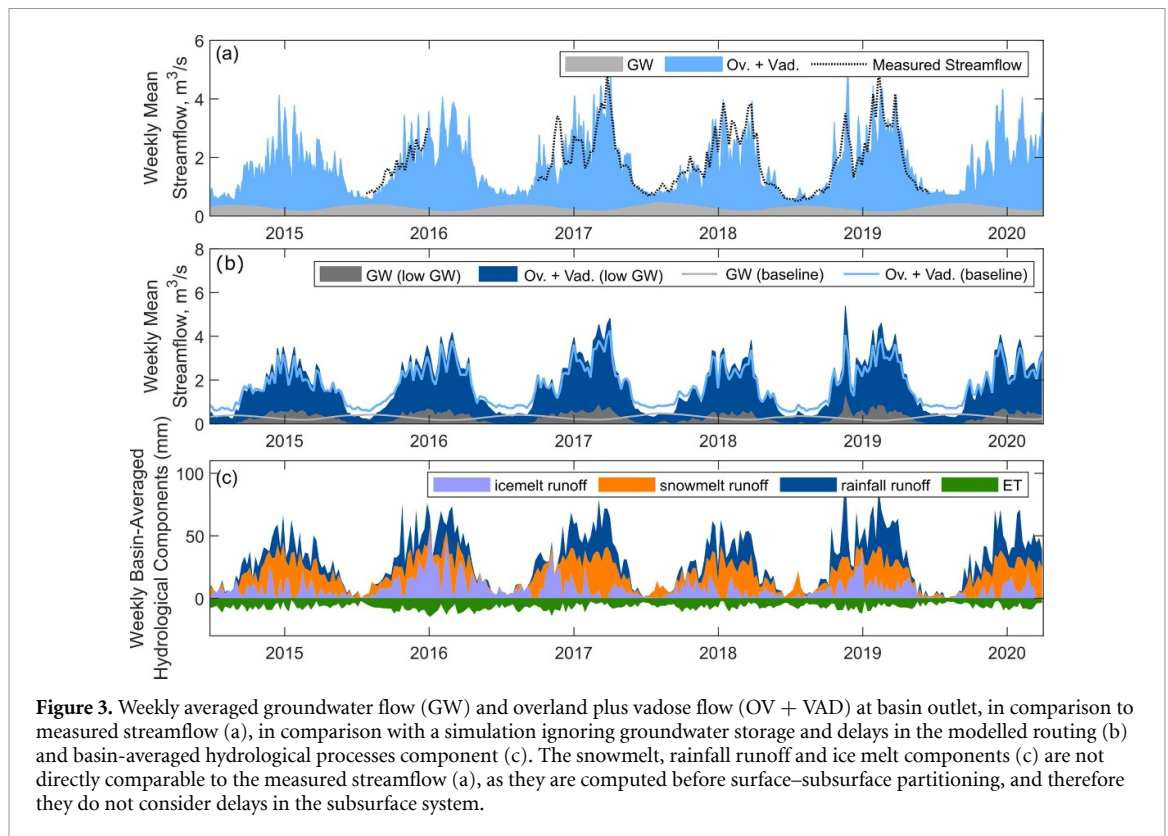


Table 3. Overland plus vadose zone flow (runoff) and groundwater contribution to streamflow at the basin outlet (%) and basin-averaged water balance (%). Numbers in parentheses are monthly fluxes (mm/month).

Period	Streamflow components		Water balance			
	Overland plus vadose	Groundwater	Rainfall runoff	Snowmelt	Ice melt	ET
Wet season	90 (1475)	10 (158)	30 (48)	32 (51)	23 (36)	15 (23)
Dry season	63 (391)	37 (232)	12 (6)	24 (12)	27 (13)	37 (18)
December	91 (1618)	9 (166)	30 (55)	33 (64)	25 (48)	13 (25)
August	44 (208)	56 (261)	3 (1)	22 (8)	29 (12)	46 (18)
Annual	84 (887)	16 (163)	27 (26)	31 (30)	23 (23)	19 (18)

storage and delays is key to capturing the hydrological regime of the basin, as shown in figure 3(b) which shows a model falsification with no GW storage (figure 3(b)).

During the wet season, rainfall-runoff, snowmelt and ice melt contributed nearly equally to streamflow discharge from the basin (figure 3(c), table 3), at 30%, 32% and 25%, respectively. During the dry season, these ratios shifted to 12%, 24% and 27%. ET fluxes, which here refer to open water evaporation, snow and ice sublimation and vegetation evapotranspiration, were similar between the dry and wet seasons due to the different controls on this process (figure

S1). However, ET is a larger relative water balance component during the dry season (37%) compared to the wet season (15%). During the wet season, more water is available for ET, but high cloud cover limits the available energy. In the dry season, water availability is reduced, leading to lower ET losses. However, the glacier surface and wetlands in the valley bottom continue to promote high evapotranspiration fluxes at a lower, but steady rate. Therefore, when comparing basin-wide evapotranspiration flux on a monthly timescale, it provides a consistent value throughout the wet and dry seasons, while daily variability and spatial variability offer larger heterogeneity, which

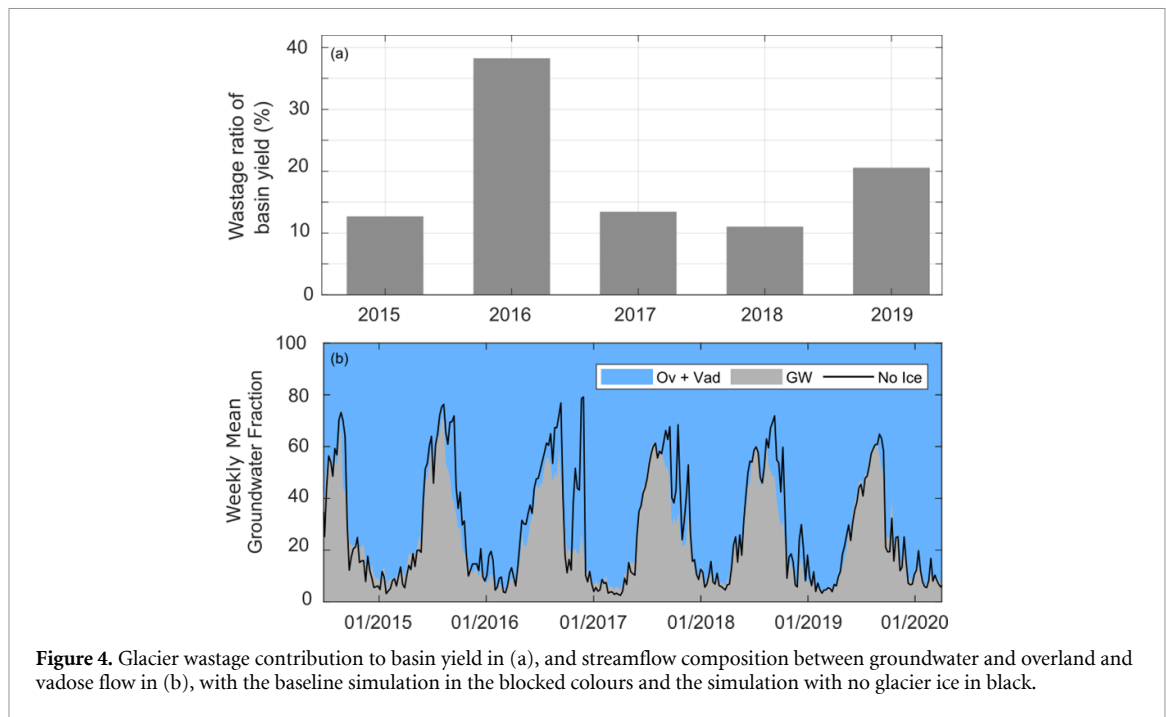


Figure 4. Glacier wastage contribution to basin yield in (a), and streamflow composition between groundwater and overland and vadose flow in (b), with the baseline simulation in the blocked colours and the simulation with no glacier ice in black.

explains the difference between the modelled ET rates from this study and the literature values (table 2).

3.3. Importance of glacier ablation in streamflow composition

At the annual scale, the imbalance melt (the glacier ablation that exceeds the annual accumulation on the glacier, also called wastage; i.e. Miles *et al* (2021), Comeau *et al* (2009), supplementary text ST4) contributes to between 10% and 38% of basin yield, with an average of 19% (figure 4(a)). In this case, basin yield is the sum of the streamflow-generating hydrological processes—rainfall runoff both on and off the glacier, snowmelt off-glacier, and both balanced and imbalanced melt. In other words, an average of 19% of the water in the basin is provided by a negative mass balance in the glacier, and is therefore a ‘non-renewable’ source of water in the basin. However, the balanced and imbalanced melt cannot be calculated either seasonally or tracked in the routing between the surface and subsurface flow paths to provide an estimate of how much GW flow originates from glacier ablation.

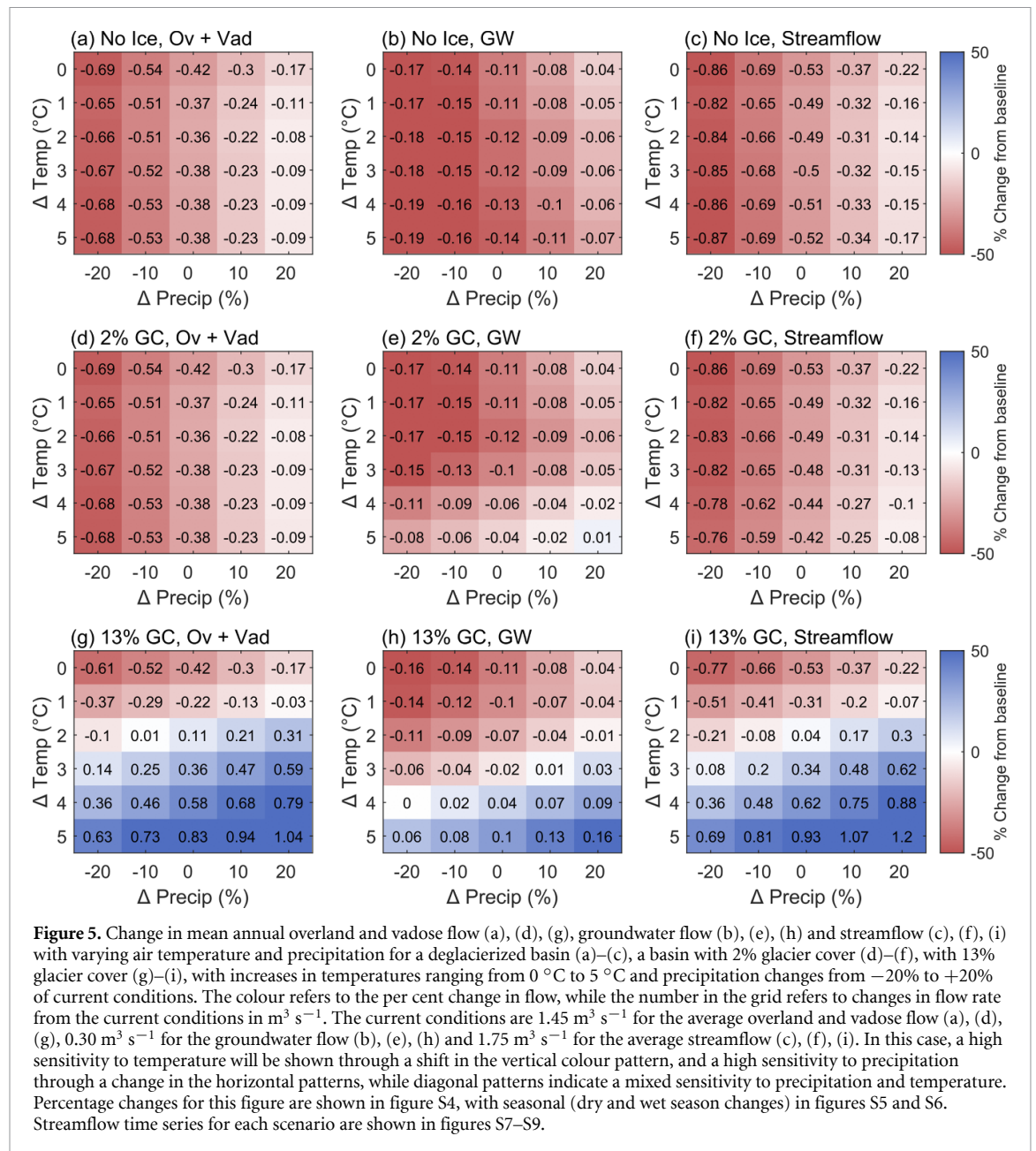
To provide an approximation of how surface and GW partitioning is influenced by melting glacier ice, a second falsification run is done without glacier coverage, with a discussion of the limitations of this approach in ST4. In the simulation without glaciers under current conditions, streamflow fraction routed through the GW reservoir shows limited changes during the wet season, but it generally increases in the September to December period (figure 4(b)). This suggests that the removal of glacier in the basin causes a larger decrease in the overland and vadose zone flow

compared to the GW, showcasing the resilience of the GW system.

3.4. Sensitivity to temperature and precipitation

In a completely deglaciaded basin, changes in air temperature had limited impacts on overland and vadose zone flow, GW flow and streamflow (figures 5(a)–(c) and S7), with larger changes linked to shifts in precipitation. On average, in a deglaciaded basin, a 20% increase in precipitation resulted in a decrease of 9%–12% of annual streamflow compared to the baseline simulation (current climate with +0 °C for temperature, 1 × precipitation and 26% glacier cover). This showcases that the increase in precipitation is unable to compensate for the total loss of glacier wastage contribution to streamflow. The decrease was more noticeable for the GW flow (–14% to –25%) than for the overland and vadose zone flow (–6% to –12%). While the largest fractional changes occurred in the GW flow, the largest volumetric changes occurred in the overland and vadose zone flow.

When the glacier cover is 2% of the basin area, located at the highest elevations, the response of streamflow and overland and vadose zone flow (figures 5(d), (f) and S8) was very similar to that of a fully deglaciaded basin. However, the GW flow showed a different response (figure 5(e)). In that case, a sensitivity to temperature change was detected, causing a larger range of variations in GW volume. For example, under an increase of 20% in precipitation, GW flow could vary between a decrease of 14% to an increase of 5% from the baseline simulation with varying temperatures, linked with increased glacier melt under high temperatures. In this case, the leftover glacier ice was able to enhance the precipitation change

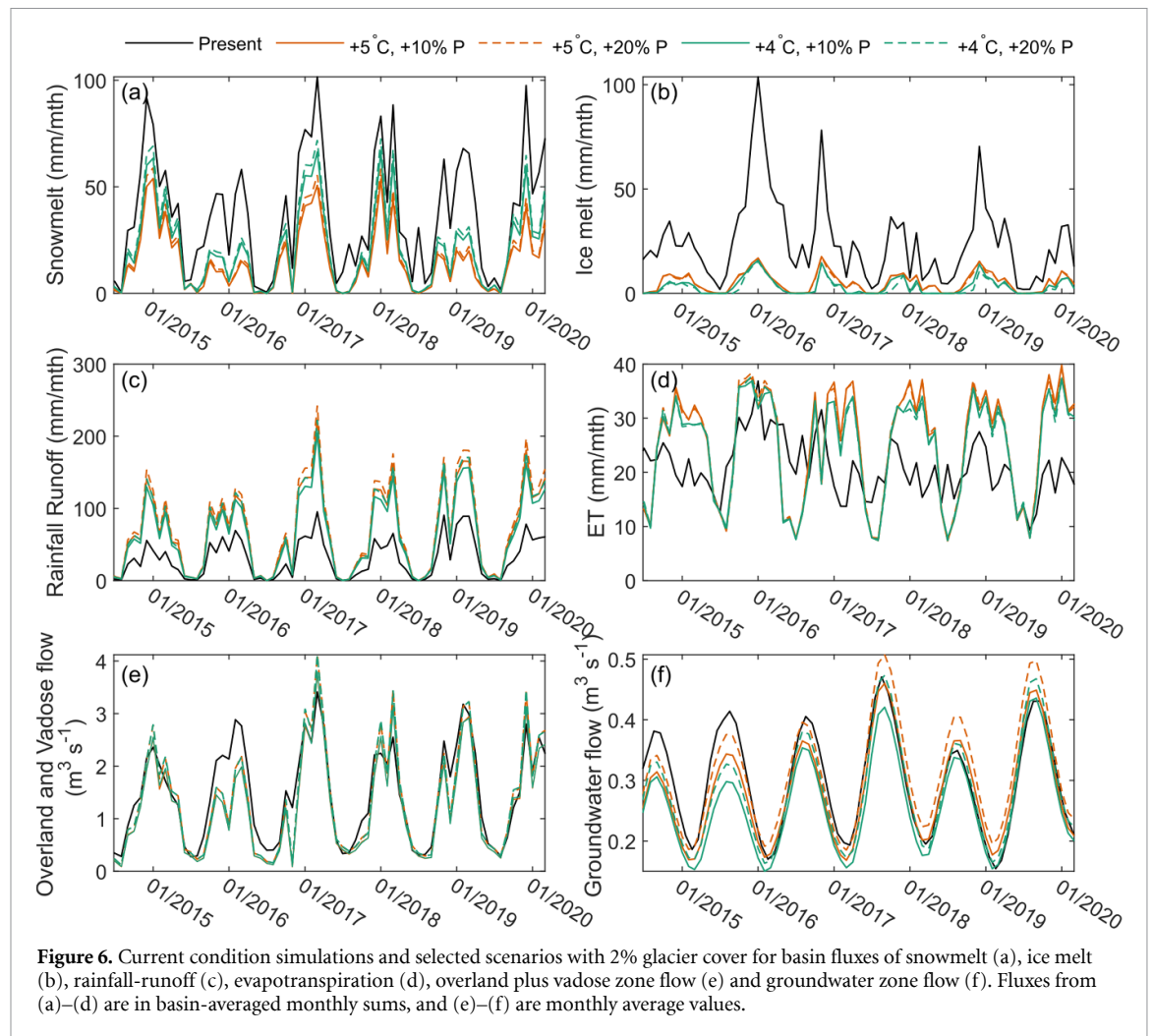


under warm temperature and sustain the GW flow component, if not the overland and vadose zone flow.

The largest variability in streamflow response was associated with a glacier coverage of 13% (figures 5(g)–(i) and S9). Under these conditions, the basin streamflow discharge was very sensitive to both temperature and precipitation changes, and these sensitivities could dampen the influence of glacier retreat under certain scenarios. For example, with an increase in temperature of 2 °C, and no change in precipitation, annual streamflow would increase by 2% from baseline conditions, despite a decrease in GW contribution to streamflow (figures 5(h) and (i)). In this case, the enhanced melt from the leftover glacier ice compensates for the partial glacier loss in the basin and sustains streamflow, without a need for

increased precipitation. This 13% glacier cover, in the range of a future climate under scenarios SS1-2.6 or RCP 2.6, is in line with a mitigated climate and global median temperature change below 2.0 °C warming relative to 1850–1900, with implied net zero emissions in the second half of the century (Lee *et al* 2021; figure ST2.4), and therefore aligns with these lower magnitude changes in temperature and precipitation.

The combination of precipitation and temperature change under a 13% glacier cover also showcases how variable the streamflow response can be. Under 13% glacier cover, an increase in precipitation of 20% combined with an increase in 5 °C would increase both overland and vadose zone flow and GW flow by more than 50%, with the opposite scenario (0 °C and –20%) resulting in a decrease in the flow components by 42%–55% (figures 5(g) and (h)). This



variability is due to the strong temperature sensitivity of the remaining glaciers within the basin. Currently, the areas in the upper region of the glacier are not melting as they are located above the equilibrium line. However, with an increase in temperature, these areas transition to rapid ablation, increasing streamflow. It is key to note that while these simulations of high glacier cover, high temperature and low precipitations can help to understand the connections between glacier melt and meteorological conditions, these combinations are physically unlikely, as these meteorological conditions would not be compatible with a continuous glacier cover. These simulations should be taken as a thought experiment on basin behaviour and sensitivity to meteorological inputs and emphasize the importance of accurate meteorological forcings.

Based on the existing literature (Bradley *et al* 2004, Solman 2013, Almazroui *et al* 2021, Potter *et al* 2023, Rounce *et al* 2023), the most likely future climate scenarios are an increase in temperature of 4 °C–5 °C combined with an increase in precipitation of 9%–20%, and a glacier coverage of 0%–2%. Under these conditions, annual streamflow would decrease

by 4%–20%, with the overland and vadose flow component decreasing by 6%–16%, and a GW flow ranging from a decrease of 36% to an increase of 5%. These changes are more noticeable for the dry season, as shown in figures S7–S9.

Under these conditions, snowmelt and ice melt decrease substantially from current conditions (figures 6(a) and (b)), as more precipitation falls as rain, causing significantly lower snowpack accumulation in the basin (figure S10). This is amplified by the decrease in glacier area in the basin. Due to the combination of increased precipitation and the increased rainfall ratio (figure S11), linked to warmer temperatures, the rainfall-runoff increases (figure 6(c)). These three mass fluxes retain their seasonal patterns. Basin-averaged evapotranspiration increases but shifts in seasonality (figure 6(d)). Under current conditions, evapotranspiration slightly increases during the dry season, as glacier melt and sublimation occur during those periods. In scenarios with limited glacier and snow cover, the evapotranspiration is driven by transpiration from vegetation, and thus becomes more tightly coupled to the wet season. Due to the large uncertainties on how vegetation might colonize the

deglacierized areas, especially on how it might influence plant characteristics relevant to hydrology processes, vegetation is kept constant between the simulations. Therefore, changes in ET are only linked to energy and water availability. Further research should be conducted on the influence of mountain greening on hydrological processes to be able to include its influence in the basin-scale modelling approach as presented here.

On average, the overland and vadose zone contribution to streamflow, as well as the GW contributions to streamflow show limited changes across the simulated period (figures 5(e) and (f)), except for the year 2016, when glacier wastage contributed a large percentage of the flow composition, and therefore, the decrease in streamflow under low glacier cover is more noticeable. The overall limited changes in streamflow are due to the balance between the decrease in glacier wastage, which averaged 19% of flow under the baseline simulation, and the increase in both precipitation and evapotranspiration in the climate change scenarios. This suggests the resilience of the streamflow to shifts in glacier and climate in this alpine valley. The fine balance between precipitation changes and glacier wastage also emphasizes the importance of obtaining reliable predictions of precipitation, as this can dictate the change in water availability in mountain basins.

3.5. From conceptual understanding to hydrological model: advances and ongoing uncertainties

There remain substantial gaps in understanding mountain GW–cryosphere interactions and even greater uncertainties when trying to assess these interactions in the face of a changing climate (van Tiel *et al* 2024). Current high mountain hydrological studies with a focus on GW both emphasize the need for fully-coupled surface water and GW, high-resolution numerical models, while simultaneously debating the capacity to properly apply these models in high alpine catchment due to the lack of information. Specifically, bedrock hydrologic data essential for these models, such as permeability, porosity, and flow rates, are often extremely rare, limiting the full utilization of these models in steep, mountainous basins (Carroll *et al* 2020, 2024, Thornton *et al* 2022). This debate is occurring concomitantly with the urgent need for further information about how water resources are changing in these landscapes to guide integrated adaptation and mitigation measures (Vuille *et al* 2018, Adler *et al* 2022, Drenkhan *et al* 2023). This study demonstrates that it is possible to reconcile these opposing needs, by showcasing how a process-based hydrological model, without a full 3D numerical GW representation, can be applied to gain insights into the surface water–GW interactions of a mountain glacierized basin when relying

on a well-developed, robust conceptual understanding. This robust understanding is based on multiple shallow piezometers time series, streamflow, meteorological observations, and a unique dataset for tropical, glacierized hydrology (Mateo *et al* 2022). Building upon this intensive, long-term, monitoring, multiple process-based studies have been completed (Gordon *et al* 2015, Somers *et al* 2016) in addition to a combination of hydrochemistry and geophysical approach to derive subsurface characteristics and surface water–GW mixing in the valley (Chavez 2013, Glas *et al* 2018). These studies provided a strong physical understanding of the water flow and storage in the valley which was transferable to the modelling framework applied in the study.

However, this approach relies on multiple simplifications. The routing of GW leakage (mountain block recharge and shallow flow paths bypassing the basin outlet) is uncertain, but this is mitigated by a sensitivity analysis of various leakage scenarios (ST3.10). Furthermore, the direct calculation of how glacier wastage contributes to GW flow is limited by the model structure, as in other studies (e.g. Somers *et al* 2019). In this study, the importance of glacier melt to GW flow is assessed through a combination of calculating wastage contribution to annual streamflow, which does not directly account for surface or subsurface flow pathways, and by comparing surface–subsurface flow composition with and without glaciers, which does not differentiate between seasonal melt and wastage. Further model development to allow the tracking of source water to the subsurface reservoir is needed to allow for a further detailed investigation of cryosphere–GW interactions.

We present two significant advances for the field of mountain hydrology: first, estimates of streamflow change that include both physically-based cryosphere and GW processes are crucial in a rapidly changing environment where water demands and availability are a key concern in the upcoming decades. Second, for the wider hydrological mountain community, this study provides a framework of how middle-ground approaches can be used to address gaps in our understanding of the hydrological system by combining conceptual understanding of a system with process-based hydrological modelling. We suggest that this conceptual understanding can be developed and applied in other mountain regions by designing field experiments that provide information, both in the form of hydrological parameters and conceptual understanding, that can be integrated into the hydrological model structure. For example, hydro-chemical mixing experiments can provide estimates of GW contribution to streamflow in different parts of the watershed which can constrain hydrological models both qualitatively and quantitatively even when time-series GW head data is unavailable, and/or a

3D model is unfeasible. Furthermore, shallow observation wells (mechanically drilled or even hand-augured) can provide some constraining information on GW dynamics and parameters where deeper wells are not possible, which can improve GW representation in mountain hydrological models as shown in this study. The resulting hydrological models can then be used to develop locally applicable estimates of future water that include both GW and cryosphere processes. This approach would be most transferable to glacierized high-mountain basins where hydro-meteorological datasets and process-based studies already exist, and where an additional few studies of GW flow path could be used to complement the dataset.

4. Conclusion

Mountain regions play a vital role in a changing climate, supplying crucial water resources to downstream communities. This study explored glacier melt, GW, and streamflow dynamics in a tropical glacierized basin. It revealed that during both the dry and wet, overland plus vadose zone flow dominates streamflow (83%), with higher GW contribution in the dry season (37%) than in the wet season (10%). Additionally, 3% of basin yield was estimated to bypass the streamflow gauge as shallow subsurface flow or mountain block recharge outside of the drainage basin.

Scenario simulations show the basin's sensitivity to temperature, precipitation, and glacier cover changes. In a low-glacier cover future, precipitation may have a greater impact on streamflow than temperature, but increased glacier coverage leads to a dual sensitivity due to glacier melt. A likely climate change scenario for the region under RCP 8.5 of a 4 °C–5 °C temperature increase with a 10%–20% precipitation increase, combined with a glacier cover of 0%–2% suggests a 4%–19% decrease in annual streamflow, coming predominantly from a decrease in the overland and vadose zone flow.

This research highlights the potential of hydrological models guided by a strong conceptual understanding of subsurface hydrology to diagnose hydrological function and sensitivity to climate change. Despite simplifications and manual calibration, the model effectively replicates observed behaviours in a tropical glacierized basin, emphasizing the importance of cryosphere–GW interactions and GW system resilience in the face of climate change.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://zenodo.org/doi/10.5281/zenodo.10870763> (Aubry-Wake 2024).

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